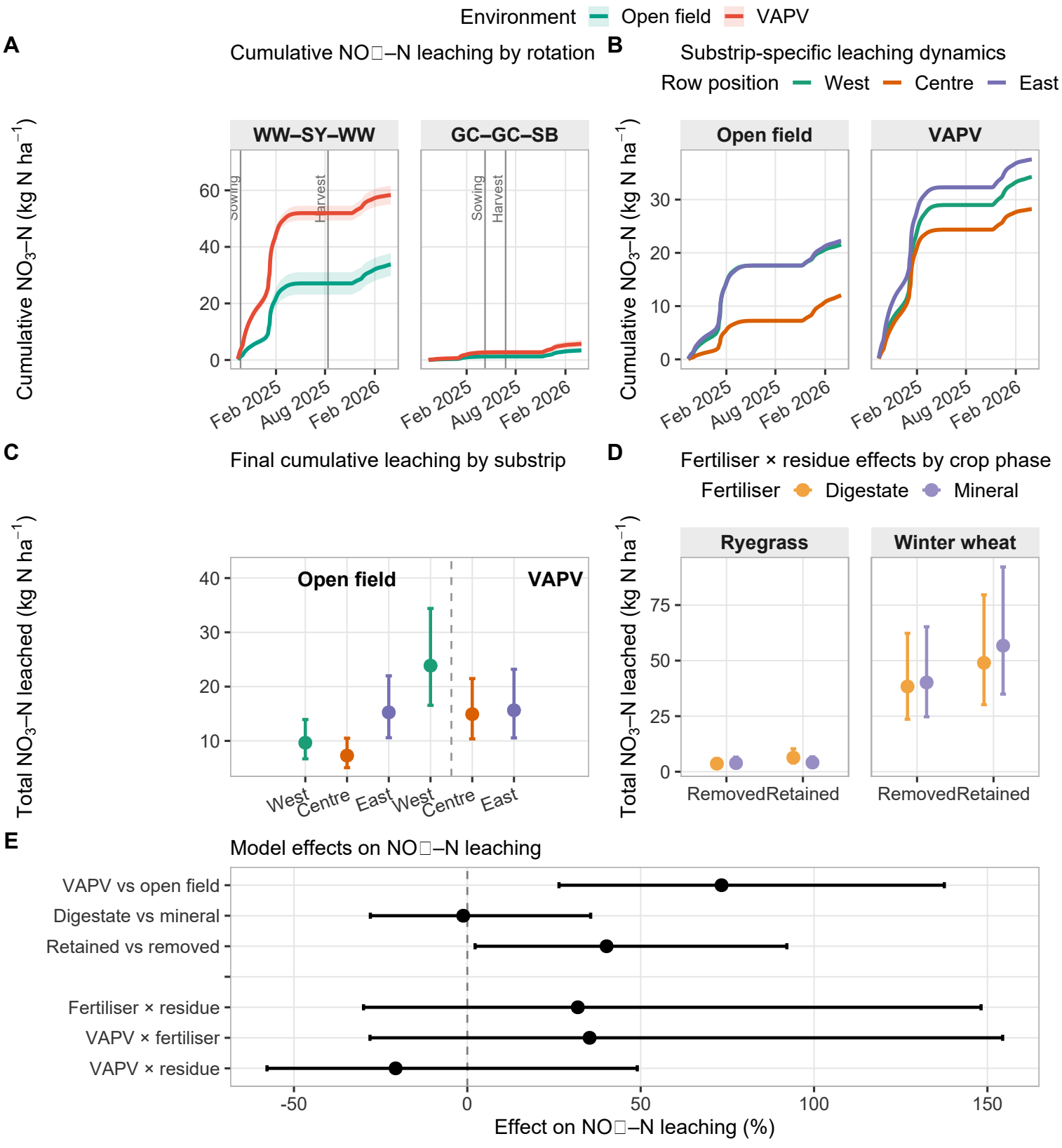

Environment
Open field
VAPV
A
Cumulative NO□–N leaching by rotation
WW–SY–WW
GC–GC–SB
Sowing
Harvest
Cumulative NO3–N (kg N ha−1)
Feb 2025
Aug 2025
Feb 2026
B
Substrip-specific leaching dynamics
Row position
West
Centre
East
Open field
VAPV
C
Final cumulative leaching by substrip
Total NO3–N leached (kg N ha−1)
West
Centre
East
D
Fertiliser × residue effects by crop phase
Fertiliser
Digestate
Mineral
Ryegrass
Winter wheat
Removed
Retained
E
Model effects on NO□–N leaching
VAPV vs open field
Digestate vs mineral
Retained vs removed
Fertiliser × residue
VAPV × fertiliser
VAPV × residue
Effect on NO□–N leaching (%)